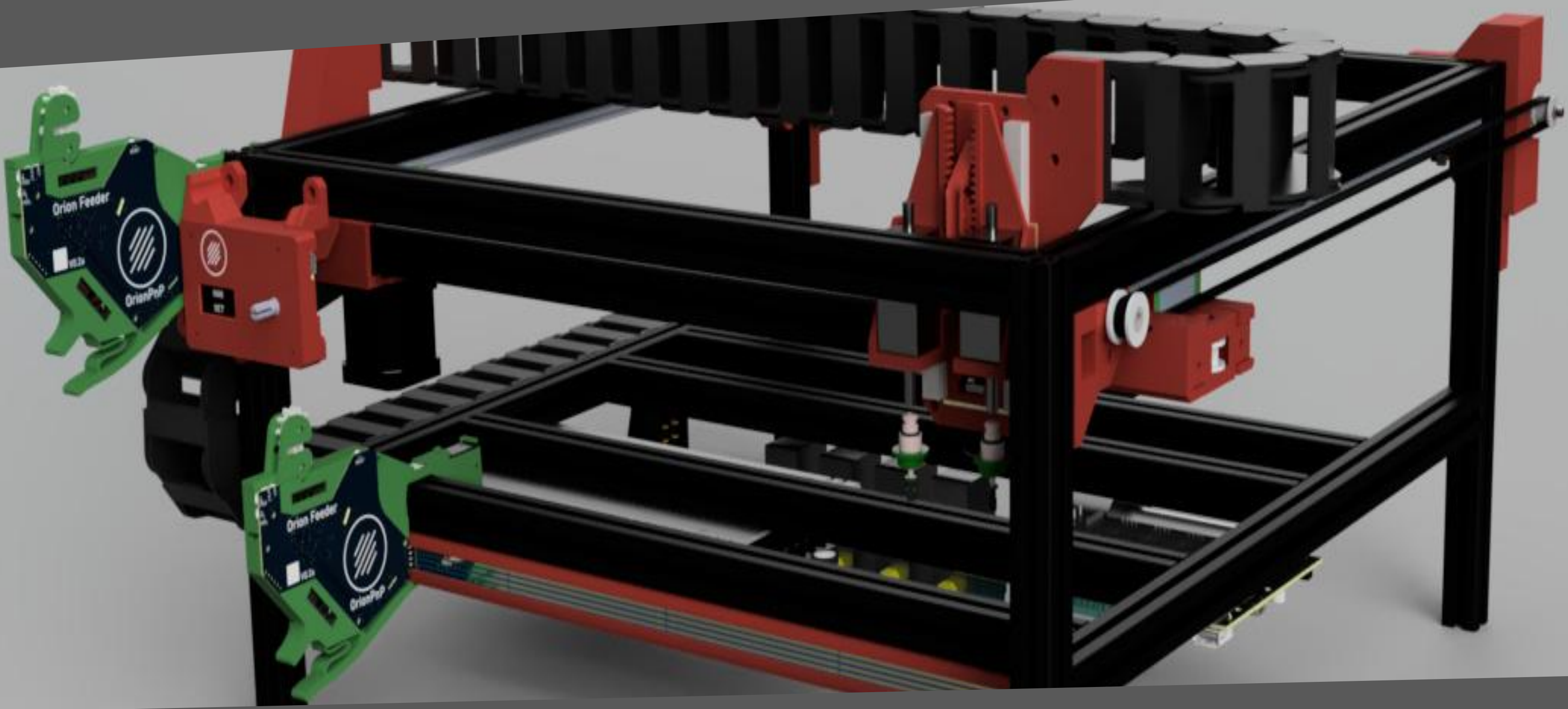
OrionPnP

Assembly manual



Rev.
01a

A word about safety

You are about to build a machine, powered by mains voltage, which could become DEADLY if not handled properly. Please consult a professional if you are unsure on how to proceed when the symbol is shown.

We are going to build a machine, capable of injuring you and/or others. There is no pride in getting hurt or worse, so be careful and always wear adequate protective gear when needed.

In the manual the symbol is used whenever special care is required. Make sure to read the manual before starting!

A word about this project

OrionPnP

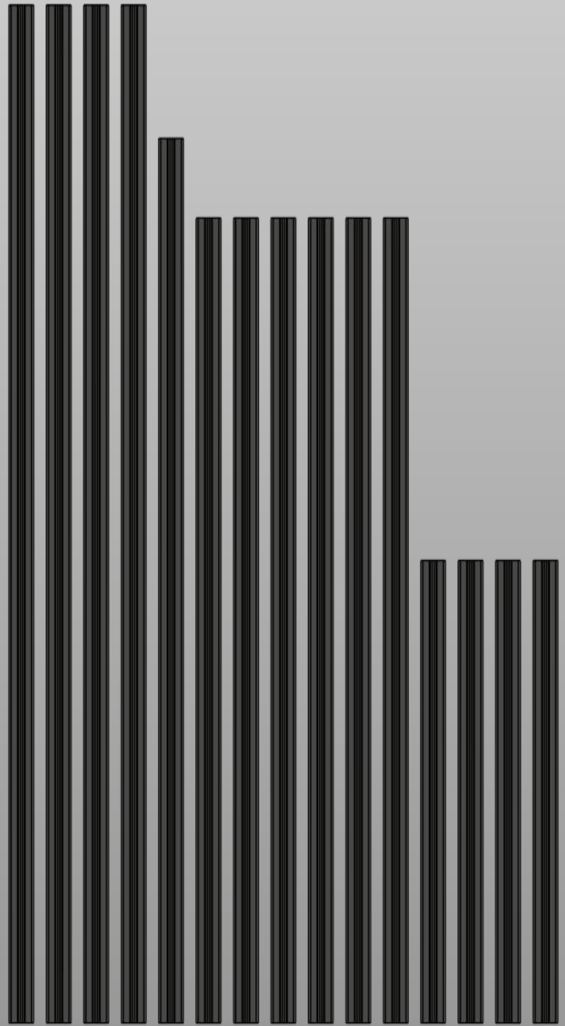
The OrionPnP project has been conceived to bring good performance pick-and-place machines to makers, with solid and reliable operation, the project aims to handle 0402 components to empower makers and allow smaller boards to be manufactured. The heart of the project is the open source software OpenPNP of course!
The project is open source, and as such every reference in this manual is made to a specific machine version. Mods, add-ons and other non official modifications will have their own guides, so follow those.
The source of each and every information is (and will always be) the Github repository.
Go and check the project if you haven't already, read the wiki and if this is the project for you, get ready to build!

How to navigate this manual

The manual

This manual will take you step by step from a bunch of components to a working OrionPnP machine. Each step is written as clearly as possible, but the drawings and the representations should clarify any doubt.
Several tools will be needed, and the list can be found in the BOM. 3D printed parts require no particular setting and can be printed in PLA, only motor mounts should be printed in heat resistant material as ABS, ASA or PETG.
Several screws will be used, make sure to reference the BOM for the codes relative to those. In the steps where different sizes will be used, a 1:1 scale image of the screws will be included as reference.
For this to be useful you must print it without scaling from the printer!

Main frame



Get the profiles ready

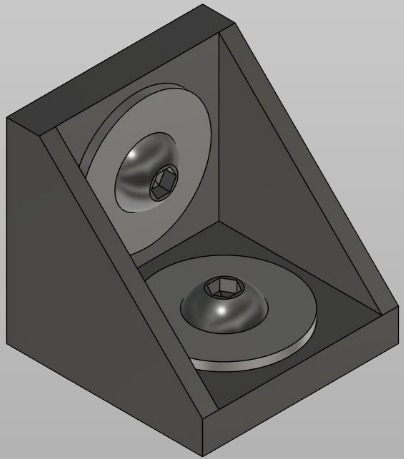
The machine is built around 2020 aluminium profiles, their sizes and amounts are:

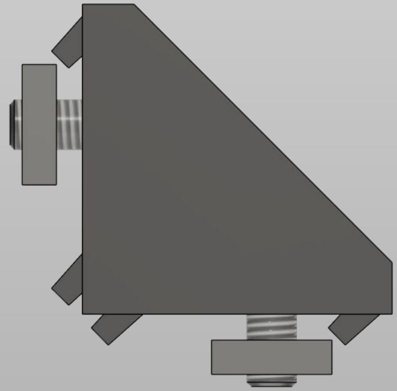
4x 250mm for the legs	6x 435mm for the X axis and bracing
1x 478mm for the X axis itself	4x 550mm for the Y axis

All these profiles need to be prepared for the next step.
No drilling or tapping is required by design, but you can tap the ends and use screws to reinforce the connections if you *really* want to.
We will start the build by assembling the “bed”, where the work surface will go later on and the rest of the frame.

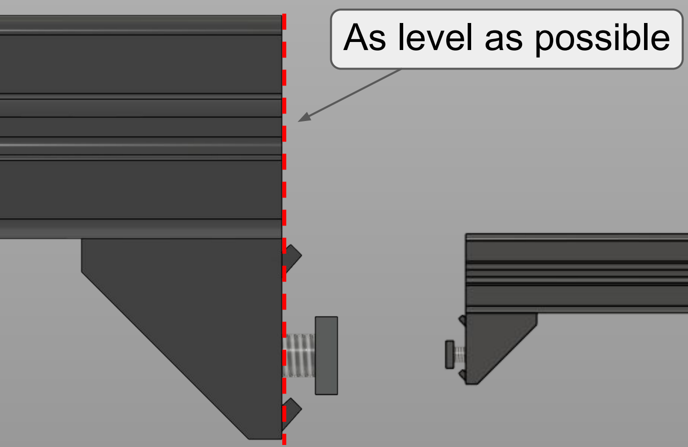
Main frame

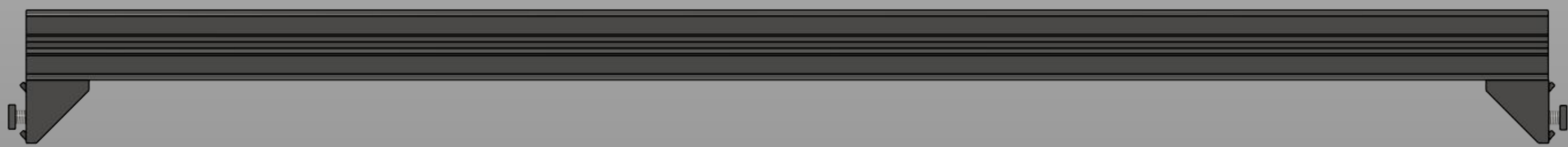
Prepare corner brackets



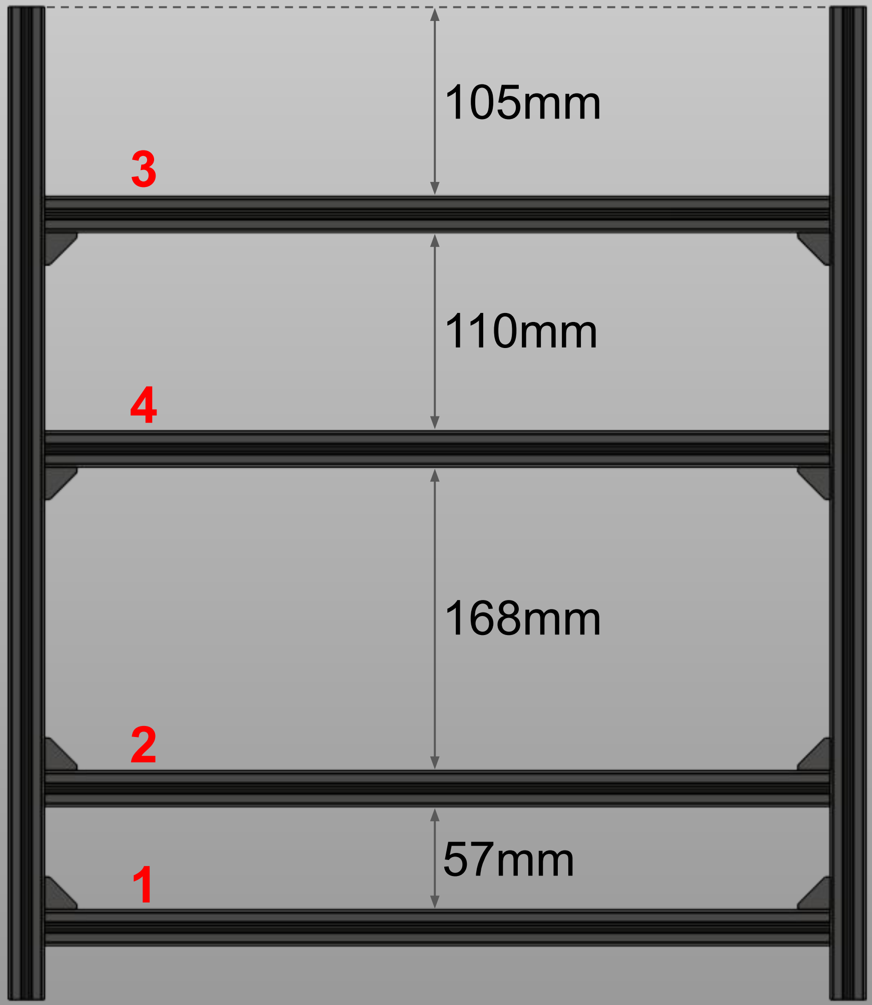


Prepare the corner brackets as illustrated, using M4x8mm button head screws and a square nut. A washer is recommended.
In total we need 20 brackets prepared this way.
Note: depending on your corner brackets you might also use M5x8mm screws.
Then prepare the 6x 435mm profiles as illustrated below. Do the same for 2x 550mm profiles. Only tighten screws on the profile side of the bracket. Leave the other nut loose!





Main frame

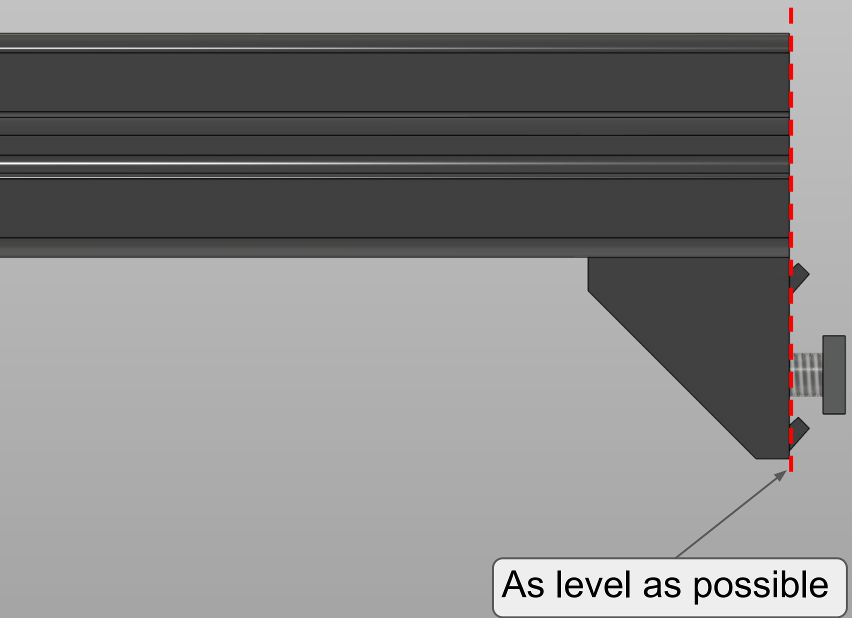


Assemble the bottom profiles

Take 4x of the 435mm profiles prepared earlier and 2x 550mm profiles.
Place them as shown in the drawing, following the order, making sure the measurements are as accurate as possible.
It is not mandatory to be 0.1mm perfect, we will calibrate in software, but try to be as precise as possible.
Pay attention to the orientation of the brackets!

***Note:** illustration is on scale so yours should look very similar to this.*

Prepare top Y beams brackets



Take the other 2x 550mm profiles and add corner brackets as shown in the picture.
Tighten only the section connected to the profile.
Then keep them to the side as we will need them later.

Now we will move on the first two 3D printed parts.

Prepare the idler holders

We will need the *Y_Corner_idler_holder_R* and *Y_Corner_idler_holder_L* files printed.

We will also need:

2x M3 heat set inserts

2x M5 heat set inserts

4x M5 square nuts

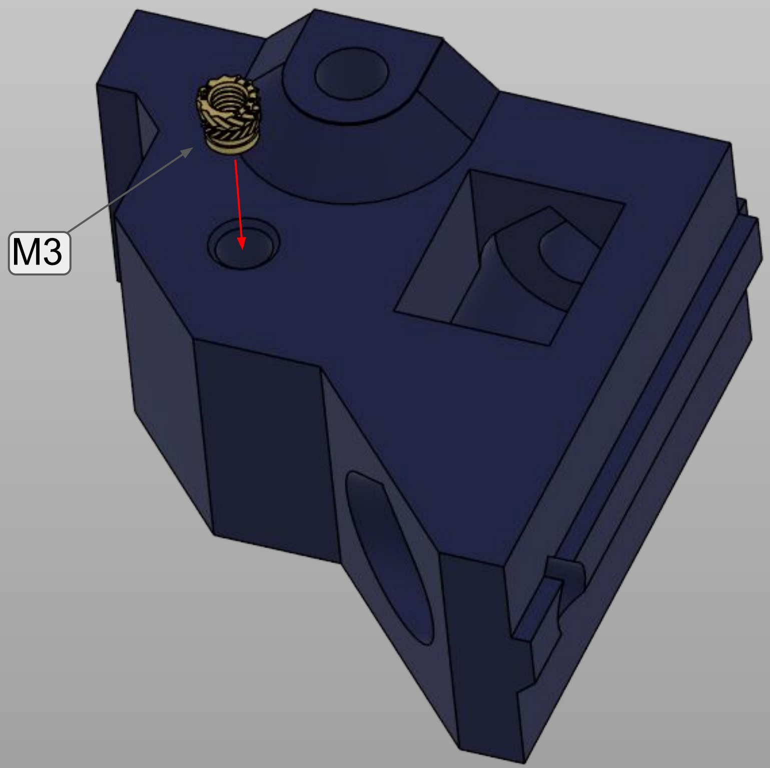
4x M5x10mm button head screws

2x M5x35mm cap head screws

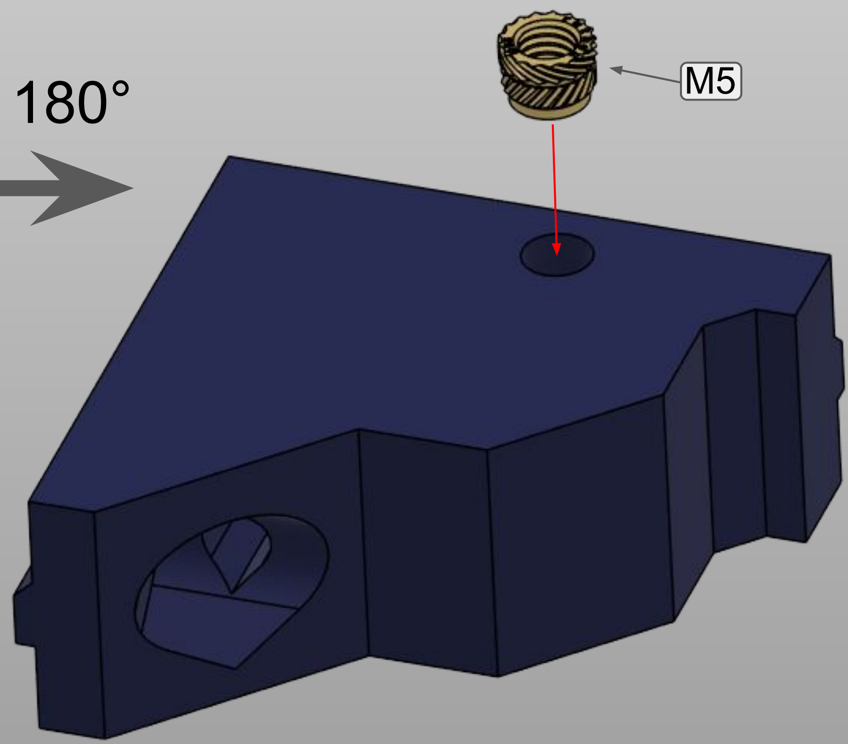
2x Toothed GT2 idlers for 6mm belts

Steps will need to be done on both R and L pieces, they are symmetrical.

Main frame

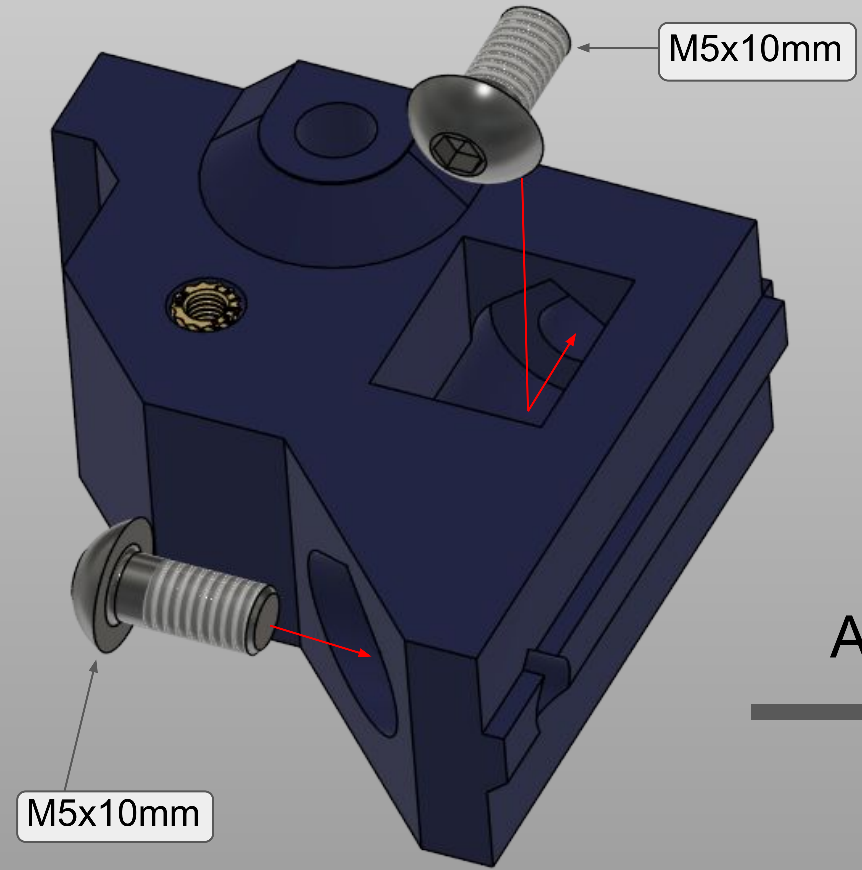


Turn the piece 180°



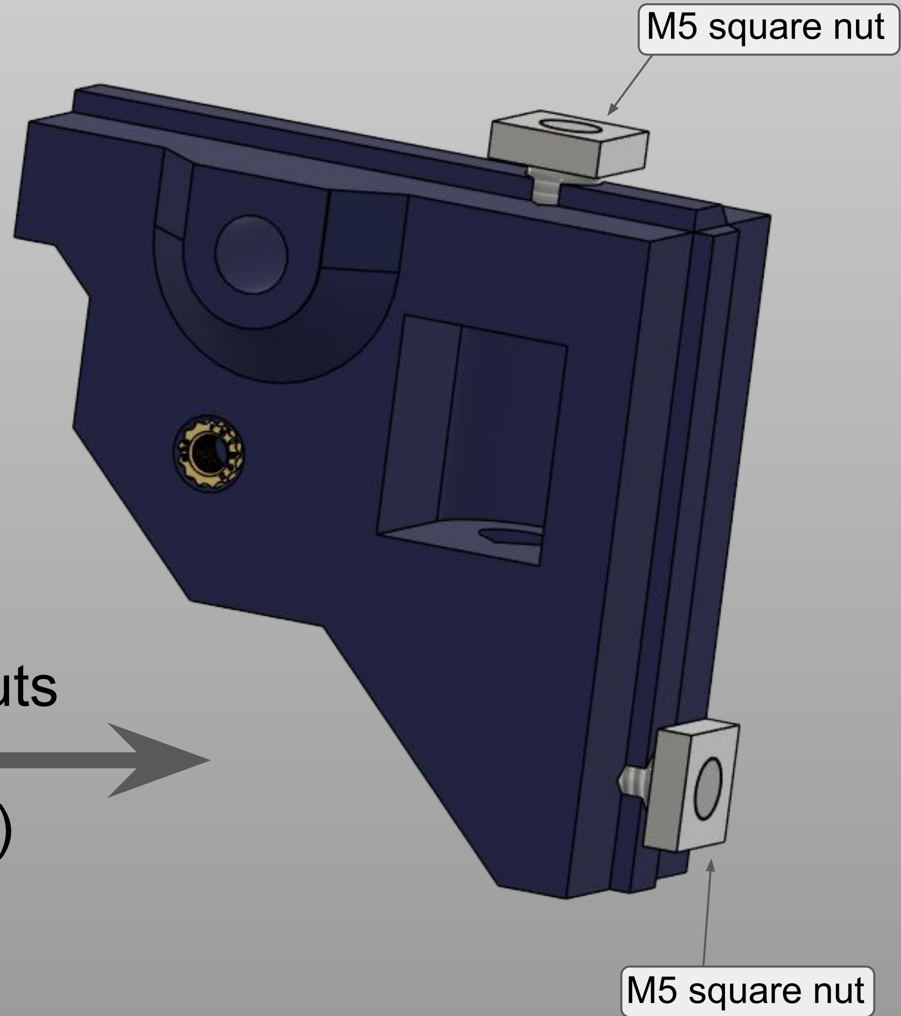
Note: *L idler holder shown in the pictures*

Main frame



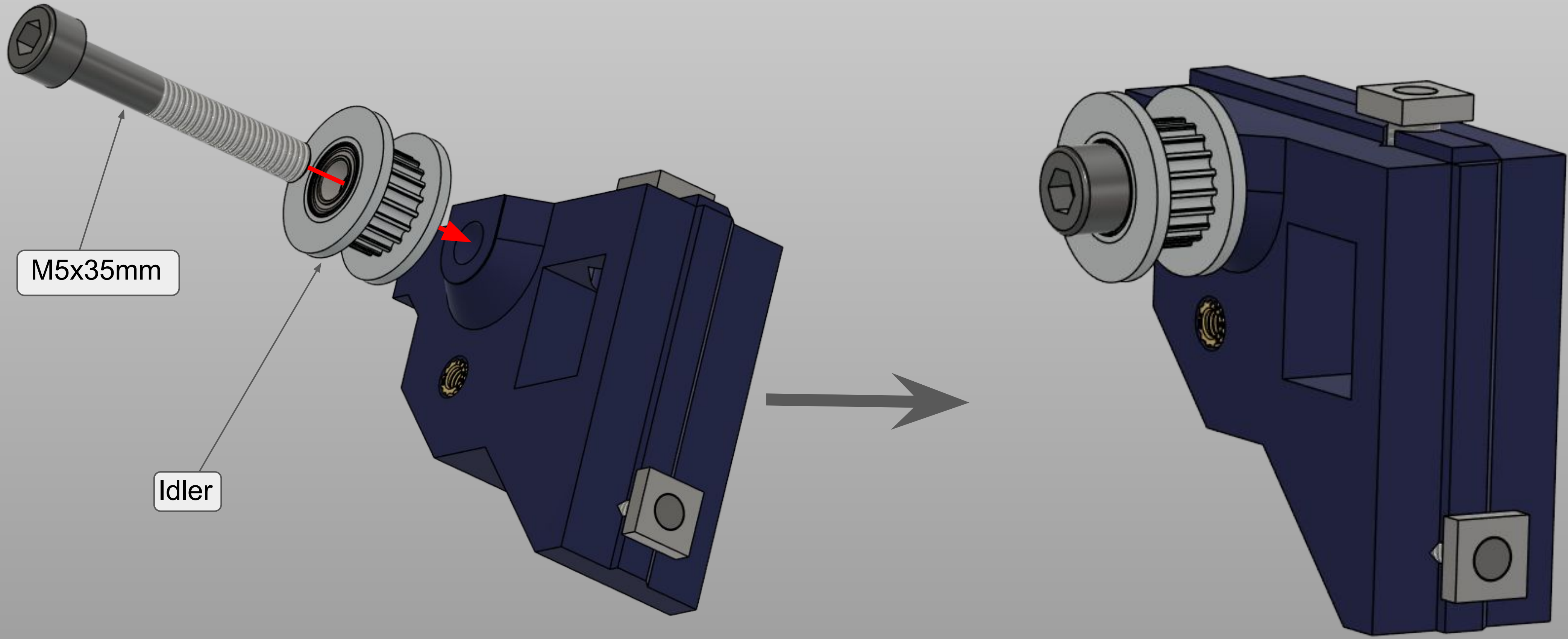
Add square nuts

(leave loose)



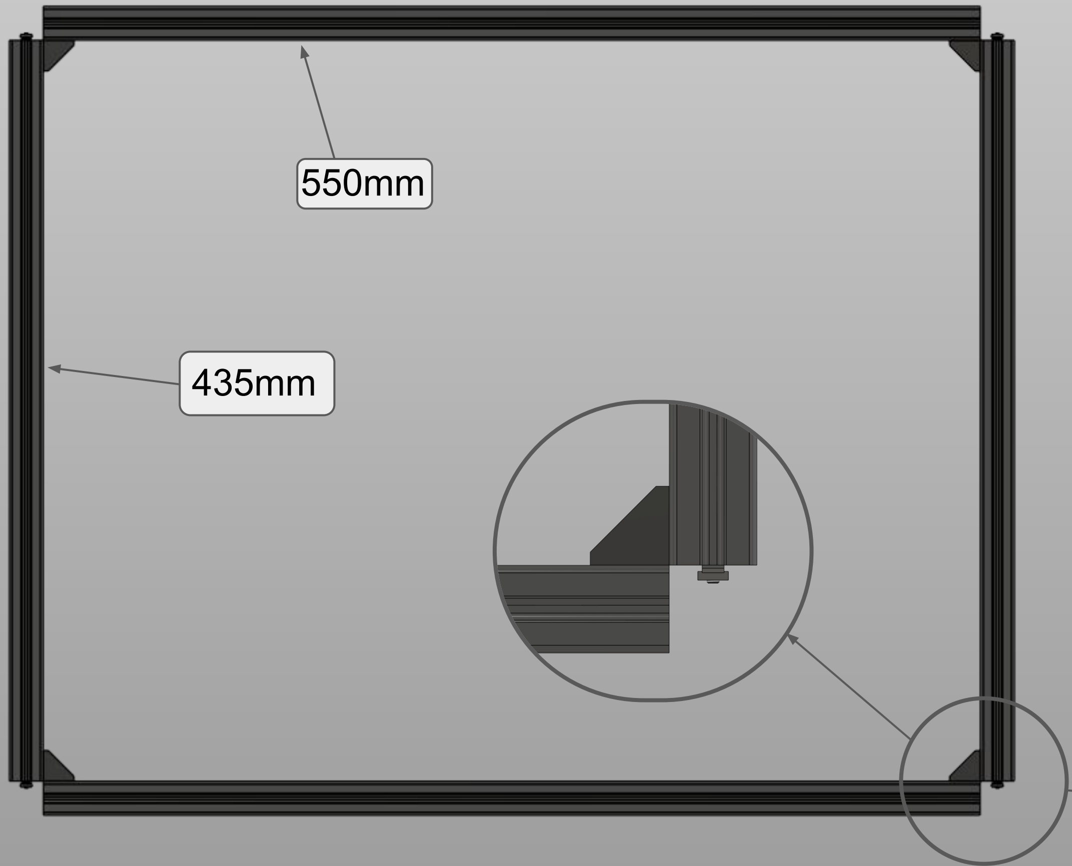
Note: *L idler holder shown in the pictures*

Main frame



Note: *L idler holder shown in the pictures*

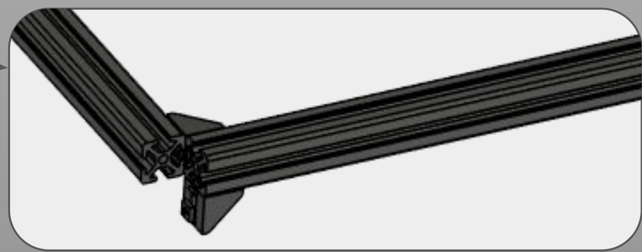
Main frame



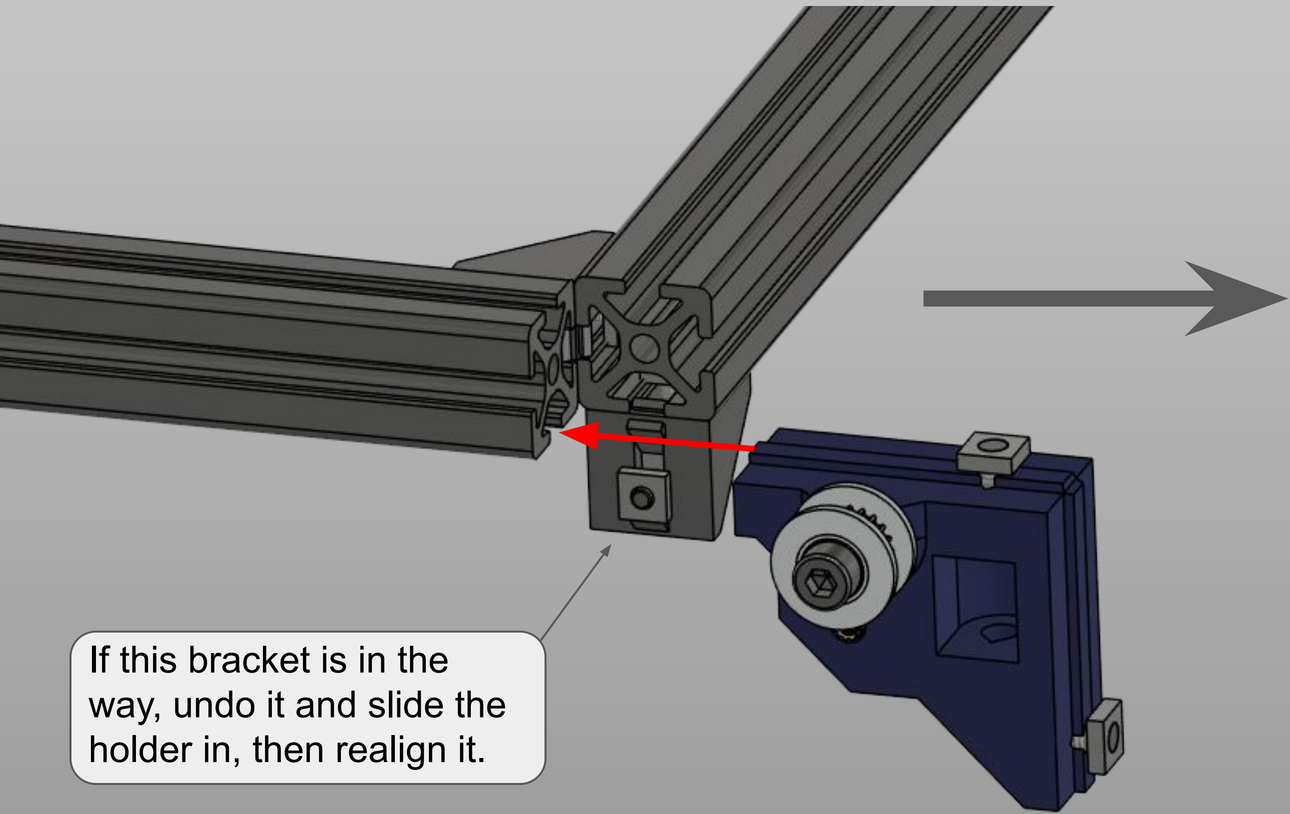
Assemble the top frame

Now take the 2x 550mm and 2x 435mm profiles prepared earlier and assemble as shown. Be careful of the orientations, and make sure the two profiles on the edge are even like shown.

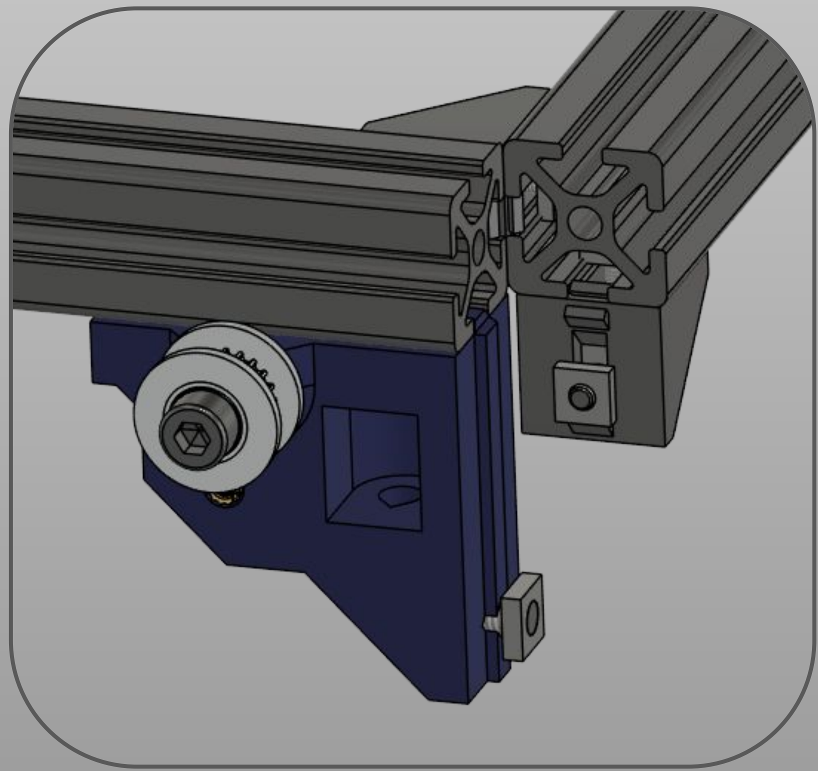
Later the leg profiles will need to fit here.



Mount the idler holders

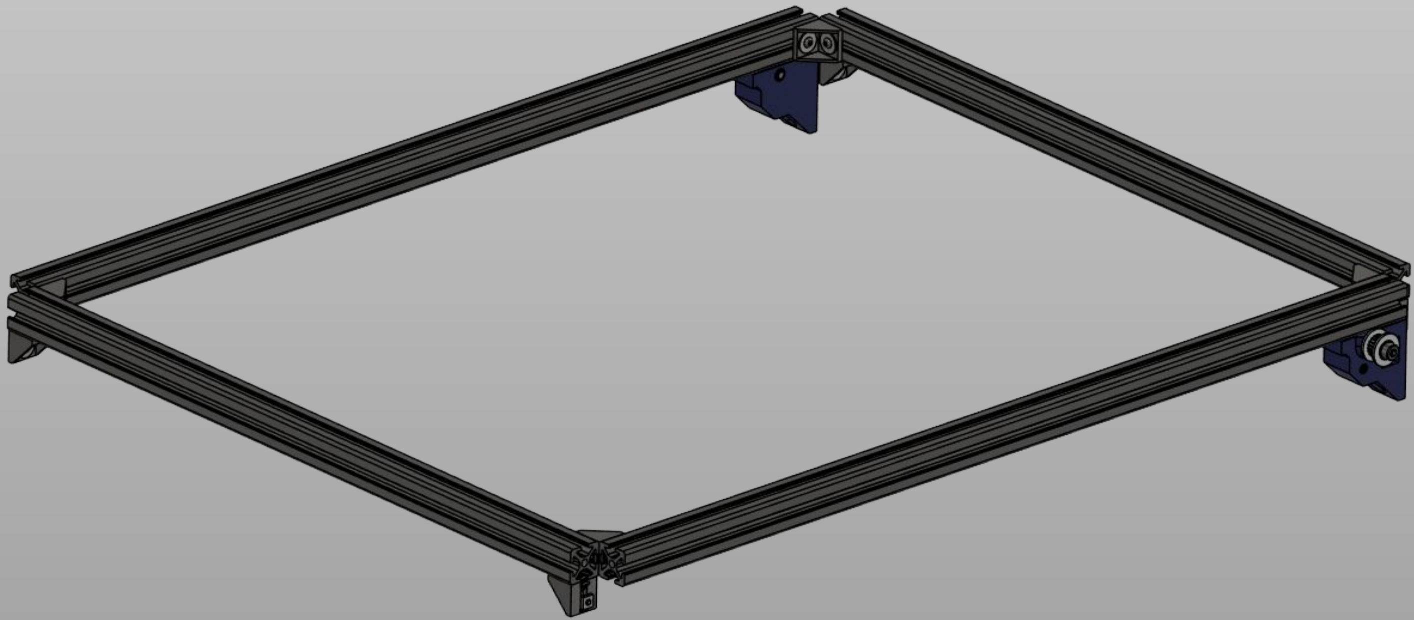


If this bracket is in the way, undo it and slide the holder in, then realign it.



Note: *L idler holder shown in the pictures*

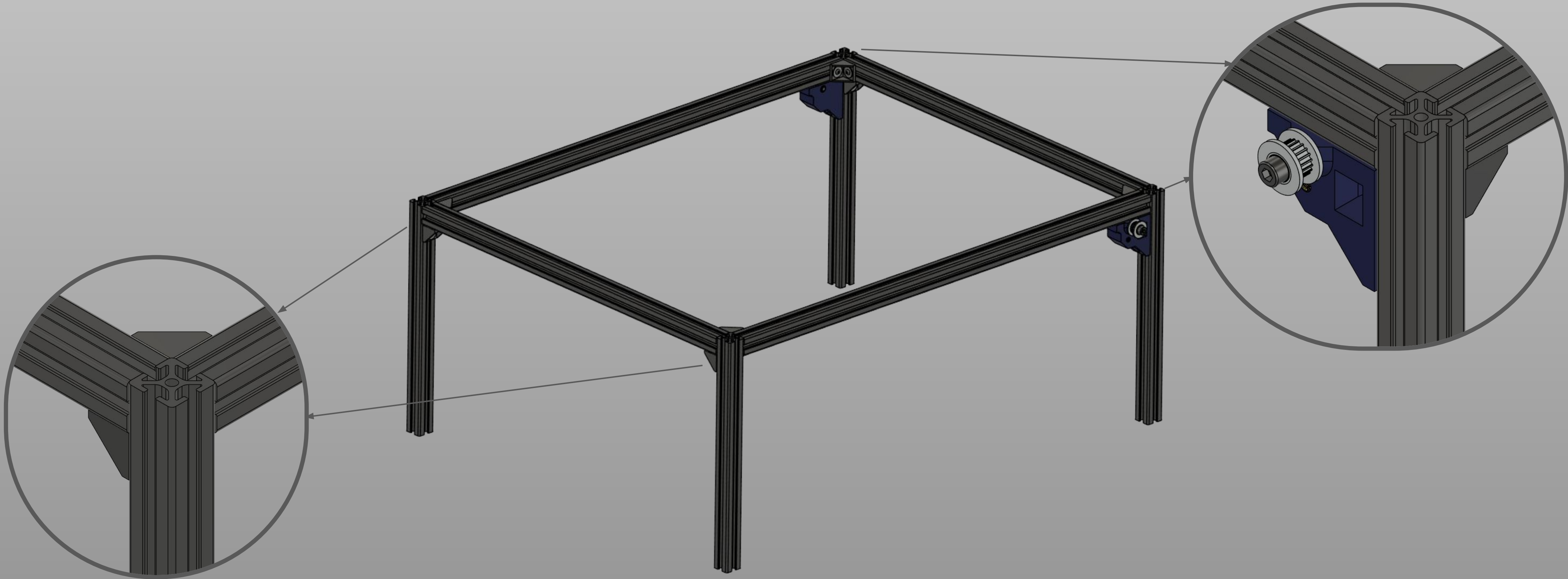
Finished top frame



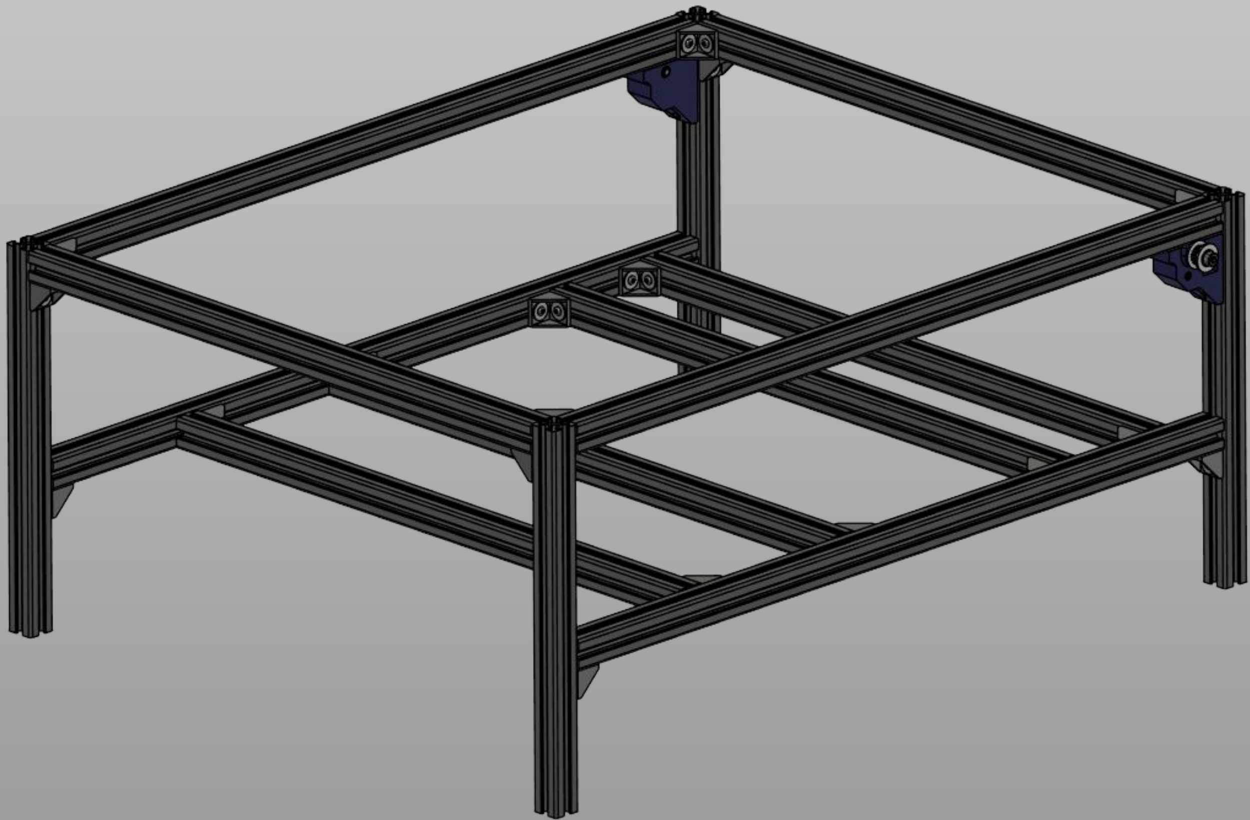
The top frame is now finished, make sure it looks like the picture shown.
Do not worry at this point if it is not aligned, we will take care in the next steps about it.

Finished top frame

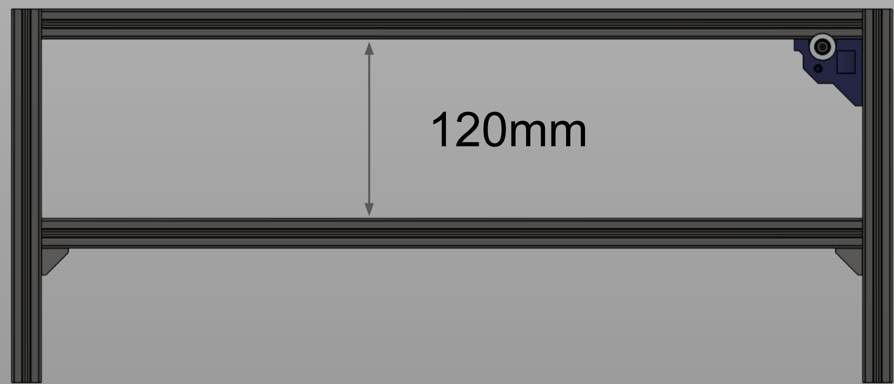
Take the 4x 250mm profiles and fix them to the top frame as shown.



Finish the frame



Now slide in the bottom frame as shown and screw the corner brackets to fix it, take care about the measurements referred below.



Square the frame

This is the time to square the frame. To do this lay the machine on the top frame, having care to check if there is any wobble.
Undo screws and fix all the profiles in their correct positions, making sure to measure the distances when needed.
If the frame is square, you should have all diagonals being equal (or close enough).
Once you are done, we can move on to the next section: rails!

Y Axis rails

Y Axis rails

Y Axis rails

Y Axis rails